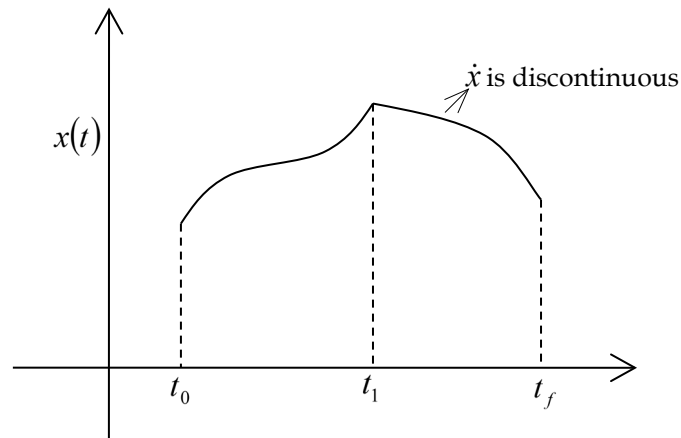


4.4 Piecewise-Smooth Extremals

$\dot{\bar{x}}(t)$ will be discontinuous except at a finite number of times in the intervals (t_0, t_f) . At a time when $\dot{\bar{x}}$ is discontinuous, $\bar{x}(t)$ is said to have a corner.



$$\dot{x}^*(t_1^-) \neq \dot{x}^*(t_1^+) \\ \text{but, } x^*(t_1^-) = x^*(t_1^+) \quad \text{🗨️}$$

t_1^- and t_1^+ denoted the times just before and just after the discontinuity $\dot{\bar{x}}$.

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt \quad \text{🗨️}$$

$x(t_0), x(t_f)$ are given, t_0 and t_f are also fixed

Let us look for necessary conditions that what must be satisfied by corner extremals (broken extremals)

$$\delta J = \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt \quad \text{🗨️}$$

Assume $x(t_1)$ and t_1 are **not related**

$$t_0 \rightarrow t_1^-, \quad t_1^+ \rightarrow t_f$$

We can apply the same method as that in free t_f case.

$$\Rightarrow \delta x_{t_1} \text{ and } \delta t_1 \text{ are free}$$

$$\delta J = \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt - \int_{t_f}^{t_1} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt$$

$\delta x_{t_1} (= \delta x(t_1) + \dot{x}^*(t_1) \delta t_1)$ & δt_1 are free

 Remember (p.33) $\delta J = \frac{\partial g}{\partial \dot{x}} \delta x_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt = 0$

Similarity we have

$$\begin{aligned} \delta J &= \left[\frac{\partial g}{\partial \dot{x}} \delta x_{t_1} + \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right) \delta t_1 \right]_{t=t_1^-} + \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &\quad - \left[\frac{\partial g}{\partial \dot{x}} \delta x_{t_1} + \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right) \delta t_1 \right]_{t=t_1^+} + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &= \left[\frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^-} - \frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^+} \right] \delta x_{t_1} + \left[\left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^-} - \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^+} \right] \delta t_1 \\ &\quad + \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &= 0 \end{aligned}$$

$\delta x(t_1)$ & δt_1 are independently arbitrary, so that their coefficients each be zero.

Necessary conditions

i) Euler-Lagrange equation

ii) $\frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^-} = \frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^+}$ 

iii) $g - \frac{\partial g}{\partial \dot{x}} \dot{x} \Big|_{t=t_1^-} = g - \frac{\partial g}{\partial \dot{x}} \dot{x} \Big|_{t=t_1^+}$

ii) & iii) are Weierstrass-Erdman Corner conditions hold at t_1 , where the extremal has a corner.

If there are several times, $t_1, t_2, \dots, t_n$ when corner exist, then at each such time those corner conditions must be satisfied.

If $\delta x(t_1)$ & δt_1 are related by $x(t_1) = \theta(t_1)$

$$\delta x_{t_1} \approx \dot{\theta}(t_1) \delta t_1$$

$$\begin{aligned}
\delta J &= \frac{\partial g}{\partial \dot{x}} \dot{\theta}(t_f) \delta t_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x}^*(t_f) \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= \left[\frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^-} - \frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^+} \right] \dot{\theta}(t_1) \delta t_1 + \left[\left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^-} - \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^+} \right] \delta t_1 \\
&\quad + \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= \left[\frac{\partial g}{\partial \dot{x}} \dot{\theta} \Big|_{t=t_1^-} + \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^-} - \frac{\partial g}{\partial \dot{x}} \dot{\theta} \Big|_{t=t_1^+} - \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^+} \right] \delta t_1 \\
&\quad + \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= 0
\end{aligned}$$

We obtain Weierstrass-Erdman **condition** as

$$\left[\frac{\partial g}{\partial \dot{x}} (\dot{\theta} - \dot{x}) + g \right]_{t=t_1^-} = \left[\frac{\partial g}{\partial \dot{x}} (\dot{\theta} - \dot{x}) + g \right]_{t=t_1^+}$$

We can extend the conditions for functionals having several independent functions.

Theorem:

If $\frac{\partial^2 g}{\partial \dot{x}^2} \neq 0$ for all $x, \dot{x}$ and t in the domain of g , then no $x(t)$ with a corner can extremize J .

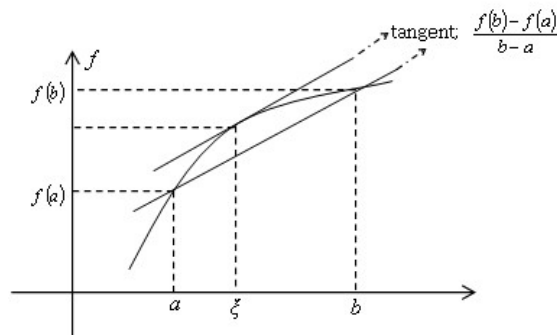
Proof.) Using mean value theorem³

³ Mean Value Theorem:

If $f(x)$ is smooth and its derivative exists in the interval (a, b) , then there exists a number ξ in (a, b) such that

$$f(b) = f(a) + f'(\xi)(b - a)$$

where $\xi = a + (b - a)\alpha$, $0 < \alpha < 1$



$$g_{\dot{x}^+} - g_{\dot{x}^-} = 0$$

$$\text{Let } f \equiv \frac{\partial g}{\partial \dot{x}}, \quad a = \dot{x}(t_1^-), \quad b = \dot{x}(t_1^+),$$

$$f(b) = \frac{\partial g}{\partial \dot{x}}(x, \dot{x}^+, t), \quad t = t_1^+$$

$$f(a) = \frac{\partial g}{\partial \dot{x}}(x, \dot{x}^-, t), \quad t = t_1^-$$

$$f' = \frac{\partial^2 g}{\partial \dot{x}^2} = g_{\dot{x}\dot{x}}$$

$$g_{\dot{x}^+} - g_{\dot{x}^-} = g_{\dot{x}\dot{x}}(x^*(t_1), \xi, t_1) [\dot{x}^*(t_1^+) - \dot{x}^*(t_1^-)]$$

$$\text{where } \xi \equiv \dot{x}^*(t_1^-) + \alpha [\dot{x}^*(t_1^+) - \dot{x}^*(t_1^-)]$$

If $g_{\dot{x}\dot{x}} \neq 0$ at any value of $x, \dot{x}, t$

Then $\dot{x}^*(t_1^+) - \dot{x}^*(t_1^-) = 0 \rightarrow$ no corner



$$\text{Example) } J = \int_{t_0}^{t_f} [a\dot{x}^2 + b x \dot{x} + c x^2] dt \quad a \neq 0, \quad x(0) = x_0, \quad x(t_f) = x_f$$

Show that $x^*(t)$ can have no corner

$$\frac{\partial^2 g}{\partial \dot{x}^2} = \frac{\partial}{\partial \dot{x}} [2a\dot{x} + bx] = 2a \neq 0 \quad (\because a \neq 0) \rightarrow \text{no corner!!!}$$

4.5 Constrained Extrema

Point(path) constraints

Let us determine a set of necessary conditions for a function x^* to be an extremal for a function of a form

$$J(x) = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt$$

subject to $f(x(t), t) = 0$

The path constraint would be present if the admissible curves were required to lie on a specified surface.

$x(t_0), x(t_f)$ are given

Form the augmented functional by adjoining the constraining relations to use Lagrange multiplier

$$J_a(x) = \int_{t_0}^{t_f} (g + \Lambda f) dt \quad \Lambda; \text{ optimal weight function } \Lambda(t)$$

When $f=0$ is satisfied, $J_a = J$

$$\begin{aligned} \delta J_a(x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} + \Lambda \frac{\partial f}{\partial x} \delta x + \delta \Lambda \cdot f \right] dt \\ &= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left\{ \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} \right] \delta x + \delta \Lambda \cdot f \right\} dt \\ &= 0 \end{aligned}$$

$$\therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} = 0$$

Euler-Lagrange equation

$$f=0$$



Differential Equation constraints

Let us find necessary conditions for a function $x^*(t)$ to be an extremal for a functional

$$J(x) = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt$$

$$\text{subject to } f(x(t), \dot{x}(t), t) = 0$$

$$x(t_0), x(t_f) \text{ are given}$$

$$J_a(x) = \int_{t_0}^{t_f} (g + \Lambda \cdot f) dt$$

$$\Lambda(t); \text{ optimal weight function}$$

$$\begin{aligned} \delta J_a(x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} + \Lambda \frac{\partial f}{\partial x} \delta x + \Lambda \frac{\partial f}{\partial \dot{x}} \delta \dot{x} + \delta \Lambda \cdot f \right] dt \\ &= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \Lambda \frac{\partial f}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left\{ \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} - \frac{d}{dt} \left(\Lambda \frac{\partial f}{\partial \dot{x}} \right) \right] \delta x + \delta \Lambda \cdot f \right\} dt \\ &= 0 \end{aligned}$$

$$\begin{aligned} \therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} + \Lambda \frac{\partial f}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} &= 0 \\ f &= 0 \end{aligned}$$

Euler-Lagrange equation

Isoperimetric constraints

We want to find $x^*(t)$ to be an extremal for a functional

$$J = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt$$

$$x(t_0), x(t_f) \text{ are given,}$$

$$\text{subject to } \int_{t_0}^{t_f} F(x(t), \dot{x}(t), t) dt = c$$

(c is specified constant. In control problems, c can be a total fuel or energy available.)

This constraint can be put into the form of differential equation constraints by defining the new variable

$$z(t) = \int_{t_0}^t F(x(t), \dot{x}(t), t) dt$$

$$z(t_0) = 0, \quad z(t_f) = c \quad z; \text{ additional state variables}$$

$$\dot{z}(t) = F(x(t), \dot{x}(t), t)$$

$$F(x(t), \dot{x}(t), t) - \dot{z}(t) \equiv f(x(t), \dot{x}(t), t, \dot{z}) = 0, \quad f; \text{ a new differential const.}$$

Augmented functional

$$J_a(x) = \int_{t_0}^{t_f} (g + \Lambda \cdot f) dt$$

$$\begin{aligned} \delta J_a(x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} + \Lambda \frac{\partial f}{\partial x} \delta x + \Lambda \frac{\partial f}{\partial \dot{x}} \delta \dot{x} + \Lambda \frac{\partial f}{\partial \dot{z}} \delta \dot{z} + \delta \Lambda \cdot f \right] dt \\ &= \frac{\partial g}{\partial x} \delta x \Big|_{t_0}^{t_f} + \Lambda \frac{\partial f}{\partial x} \delta x \Big|_{t_0}^{t_f} + \Lambda \frac{\partial f}{\partial \dot{z}} \delta z \Big|_{t_0}^{t_f} \\ &\quad + \int_{t_0}^{t_f} \left\{ \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} + \Lambda \frac{\partial f}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} \right] \delta x - \frac{d}{dt} \left(\Lambda \frac{\partial f}{\partial \dot{z}} \right) \delta z + \delta \Lambda \cdot f \right\} dt \\ &= 0 \end{aligned}$$

$$\therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} + \Lambda \frac{\partial f}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} = 0 \quad \text{Euler-Lagrange equation}$$

$$\frac{d}{dt} \left(\Lambda \frac{\partial f}{\partial \dot{z}} \right) = 0 \quad \xrightarrow{f=F-\dot{z} \rightarrow f_{\dot{z}}=-1} \quad \frac{d}{dt} (-\Lambda) = 0 \Rightarrow \Lambda(t) = \text{const.}$$

$\Rightarrow$ Lagrange multiplier is constant

$$f = 0 \rightarrow \dot{z}(t) = F(x(t), \dot{x}(t), t)$$

(These eqs. must satisfy the boundary conditions, $z(t_0) = 0, \quad z(t_f) = c$)

< Second variation & Sufficient Conditions >

- The increment of a functional

$$\Delta J \equiv J(\bar{x} + \delta\bar{x}) - J(\bar{x})$$

$$\Delta J(\bar{x}, \delta\bar{x}) = \delta J(\bar{x}, \delta\bar{x}) + g_1(\bar{x}, \delta\bar{x}) \cdot \|\delta\bar{x}\|$$

linear term in $\delta\bar{x}$

$$\text{If } \lim_{\|\delta\bar{x}\| \rightarrow 0} \{g_1(\bar{x}, \delta\bar{x})\} = 0$$

then, J is said to be differentiable on $\bar{x}$ and δJ is the variation of J evaluated for the function $\bar{x}$, δJ is the linear approximation to the difference in the functional J caused by $\bar{x}$, $\bar{x} + \delta\bar{x}$.

- The increment of a functional

$$\begin{aligned} \Delta J(\bar{x}, \delta\bar{x}) &= J(\bar{x} + \delta\bar{x}) - J(\bar{x}) \\ &= J(\bar{x}) + \frac{\partial J}{\partial \bar{x}} \Big|_{\bar{x}} \delta\bar{x} + \frac{1}{2} \frac{\partial^2 J}{\partial \bar{x}^2} \Big|_{\bar{x}} (\delta\bar{x})^2 + \dots - J(\bar{x}) \\ &= \frac{\partial J}{\partial \bar{x}} \Big|_{\bar{x}} \delta\bar{x} + \frac{1}{2} \frac{\partial^2 J}{\partial \bar{x}^2} \Big|_{\bar{x}} (\delta\bar{x})^2 + H.O.T \end{aligned}$$

$$\text{where } \frac{\partial J}{\partial \bar{x}} \Big|_{\bar{x}} \delta\bar{x} \equiv \delta J(\bar{x}, \delta\bar{x}); \text{ first variation}$$

$$\frac{1}{2} \frac{\partial^2 J}{\partial \bar{x}^2} \Big|_{\bar{x}} (\delta\bar{x})^2 \equiv \delta^2 J; \text{ second variation}$$

- $\Delta J(\bar{x}, \delta\bar{x}) = \delta J(\bar{x}, \delta\bar{x}) + \delta^2 J(\bar{x}, \delta\bar{x}) + g_2(\bar{x}, \delta\bar{x}) \cdot \|\delta\bar{x}\|^2$
linear term in $\delta\bar{x}$

$\delta^2 J$ is a quadratic functional, and if $\lim_{\|\delta\bar{x}\| \rightarrow 0} \{g_2(\bar{x}, \delta\bar{x})\} = 0$, $J(\bar{x})$ is twice

differentiable. From now on, it will be tacitly assumed that we are dealing with functionals which are twice differentiable. The second variation of such a functional is uniquely defined.



- Theorem 1 : A necessary condition for the functional $J(\bar{x}(t))$ to have a minimum for $\bar{x}(t) = \bar{x}^*(t)$ is that

$$\delta^2 J(\bar{x}(t)) \geq 0 \quad (i)$$

for $\bar{x}(t) = \bar{x}^*(t)$ and all admissible $\delta\bar{x}(t)$.

For a **maximum**, the sign $\geq$ in eq(i) is replaced by $\leq$.

Proof) By definition, we have

$$\begin{aligned} \Delta J(\bar{x}, \delta\bar{x}) &= \delta J(\bar{x}, \delta\bar{x}) + \delta^2 J(\bar{x}, \delta\bar{x}) + g_2(\bar{x}, \delta\bar{x}) \cdot \|\delta\bar{x}\|^2 \\ &\quad \text{linear term in } \delta\bar{x} \\ &= \delta J(\bar{x}, \delta\bar{x}) + \delta^2 J(\bar{x}, \delta\bar{x}) + \varepsilon \|\delta\bar{x}\|^2 \end{aligned} \quad (ii)$$

where $\varepsilon \equiv g_2(\bar{x}, \delta\bar{x})$, and $\varepsilon \rightarrow 0$ as $\|\delta\bar{x}\| \rightarrow 0$

According the fundamental theorem, $\delta J = 0$ for $\bar{x}(t) = \bar{x}^*(t)$ and all admissible $\delta\bar{x}(t)$. Hence eq(ii) becomes

$$\Delta J(\bar{x}, \delta\bar{x}) = \delta^2 J(\bar{x}, \delta\bar{x}) + \varepsilon \|\delta\bar{x}\|^2 \quad (iii)$$

Thus, for sufficiently small $\|\delta\bar{x}\|$, the sign of $\Delta J(\bar{x}, \delta\bar{x})$ will be

the same as the sign of $\delta^2 J(\bar{x}, \delta\bar{x})$.

Now suppose that $\delta^2 J(\bar{x}, \delta\bar{x}_0) < 0$ for some admissible $\delta\bar{x}_0$.

Then for any $\alpha \neq 0$, no matter how small, we have

$$\delta^2 J(\bar{x}, \alpha\delta\bar{x}_0) = \alpha^2 \delta^2 J(\bar{x}, \delta\bar{x}_0) < 0$$

Hence, eq(iii) can be made negative for arbitrary small $\|\delta\bar{x}\|$.

But this is impossible, since by hypothesis $J(\bar{x}(t))$ has a minimum for $\bar{x}(t) = \bar{x}^*(t)$, i.e.,

$$\Delta J(\bar{x}, \delta\bar{x}) = J(\bar{x}^* + \delta\bar{x}) - J(\bar{x}^*) \geq 0$$

for all sufficiently small $\|\delta\bar{x}\|$

This contradiction proves the theorem. **[end proof]**

The condition $\delta^2 J(x(t)) \geq 0$ is necessary but of course not sufficient for the functional $J(\bar{x}(t))$ to have a minimum for a given function. To obtain a sufficient condition, we introduce the following concept: We say that a quadratic functional $\delta^2 J(\delta x(t))$ defined on some normed linear space $\mathfrak{R}$ is strongly positive if there exists a constant $k > 0$ such that

$$\delta^2 J(\delta x(t)) \geq k \|\delta x\|^2 \quad \text{for all } \delta \bar{x}(t)$$



- Theorem 2 : A sufficient condition for a functional $J(\bar{x}(t))$ to have a minimum for $\bar{x}(t) = \bar{x}^*(t)$, given that the first variation $\delta J(\delta x(t))$ vanishes for $\bar{x}(t) = \bar{x}^*(t)$, is that its second variation $\delta^2 J(\delta x(t))$ be strongly positive for $\bar{x}(t) = \bar{x}^*(t)$.

Proof) For $\bar{x}(t) = \bar{x}^*(t)$, we have $\delta J(\delta x(t)) = 0$ for all admissible $\delta \bar{x}(t)$, and hence

$$\Delta J(\delta \bar{x}) = \delta^2 J(\delta \bar{x}) + \varepsilon \|\delta \bar{x}\|^2 \quad \text{where } \varepsilon \rightarrow 0 \text{ as } \|\delta \bar{x}\| \rightarrow 0.$$

Moreover, for $\bar{x}(t) = \bar{x}^*(t)$,

$$\delta^2 J(\delta x(t)) \geq k \|\delta x\|^2 \quad \text{where } k = \text{constant} > 0.$$

Thus, for sufficiently small ε_1 , $|\varepsilon| < \frac{1}{2}k$ if $\|\delta x\| < \varepsilon_1$. It follows that

$$\Delta J(\delta \bar{x}) = \delta^2 J(\delta \bar{x}) + \varepsilon \|\delta \bar{x}\|^2 > \frac{1}{2}k \|\delta x\|^2 > 0$$

if $\|\delta x\| < \varepsilon_1$, i.e., $J(\bar{x}(t))$ has a minimum for

$\bar{x}(t) = \bar{x}^*(t)$, as asserted. [end proof]



- $J = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt$ $x(t_0), x(t_f)$ are given

$$\Delta J = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt$$

$$\begin{aligned}
&= \int_{t_0}^{t_f} \left\{ g(x^*, \dot{x}^*, t) + \frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x} \right. \\
&\quad \left. + \frac{1}{2} \left[\frac{\partial^2 g}{\partial x^2} \Big|_{x^*, \dot{x}^*} (\delta x)^2 + 2 \frac{\partial^2 g}{\partial x \partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta x \delta \dot{x} + \frac{\partial^2 g}{\partial \dot{x}^2} \Big|_{x^*, \dot{x}^*} (\delta \dot{x})^2 \right] + H.O.T \right\} dt \\
&\quad - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) \\
\delta^2 J &= \int_{t_0}^{t_f} \frac{1}{2} \left[g_{xx} (\delta x)^2 + 2 g_{x\dot{x}} \delta x \delta \dot{x} + g_{\dot{x}\dot{x}} (\delta \dot{x})^2 \right] dt
\end{aligned}$$

The necessary condition for a minimum (maximum) is $\delta^2 J \geq 0$ ($\delta^2 J \leq 0$) for all variations that are admissible $\delta x(t)$ must be continuous and must have continuous first derivative (weak conditions).

i.e.,	minimum $J \rightarrow \delta J = 0, \delta^2 J \geq 0$
	maximum $J \rightarrow \delta J = 0, \delta^2 J \leq 0$



< Accessory problem >

Note that $\delta^2 J = \int_{t_0}^{t_f} w(\delta x, \delta \dot{x}, t) dt$

$$\text{where } w = \frac{1}{2} [g_{xx} (\delta x)^2 + 2 g_{x\dot{x}} \delta x \delta \dot{x} + g_{\dot{x}\dot{x}} (\delta \dot{x})^2]$$

$$\delta x(t_0) = \delta x(t_f) = 0, \delta x(t) \text{ is arbitrary}$$

The problem of **extremizing** $\delta^2 J$ with respect to δx said to be the accessory problem.

Necessary condition

$$\frac{d}{dt} \left(\frac{\partial w}{\partial (\delta \dot{x})} \right) - \frac{\partial w}{\partial (\delta x)} = 0 \quad ; \text{ Jacobi differential equation}$$

Let the solution to this be $\delta v(t)$

The Jacobian equation is $(\delta x \rightarrow \delta v, \delta \dot{x} \rightarrow \delta \dot{v})$

$$\begin{aligned}
&\rightarrow \frac{d}{dt}[g_{\ddot{x}}\delta\dot{v} + g_{\ddot{x}}\delta v] - (g_{\ddot{x}}\delta\dot{v} + g_{\ddot{x}}\delta v) = 0 \\
&\rightarrow \frac{d}{dt}(g_{\ddot{x}}\delta\dot{v}) + \frac{d}{dt}(g_{\ddot{x}})\delta v + g_{\ddot{x}}\delta\dot{v} - (g_{\ddot{x}}\delta\dot{v} + g_{\ddot{x}}\delta v) = 0 \\
&\rightarrow \frac{d}{dt}(g_{\ddot{x}}\delta\dot{v}) + \left[\frac{d}{dt}(g_{\ddot{x}}) - g_{\ddot{x}}\right]\delta v = 0 \quad \textcircled{1}
\end{aligned}$$

$$\begin{aligned}
2\delta^2 J &= \int_{t_0}^{t_f} \left[g_{\ddot{x}}(\delta\dot{x})^2 + g_{\ddot{x}}\frac{d}{dt}(\delta x)^2 + g_{\ddot{x}}(\delta x)^2 \right] dt \\
&= g_{\ddot{x}}\delta\dot{x}\delta x \Big|_{t_0}^{t_f} + g_{\ddot{x}}(\delta x)^2 \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \left[\frac{d}{dt}(g_{\ddot{x}}\delta\dot{x})\delta x - \left(g_{\ddot{x}} - \frac{d}{dt}g_{\ddot{x}}\right)(\delta x)^2 \right] dt \\
\delta^2 J &= -\frac{1}{2} \int_{t_0}^{t_f} \delta x \left[\frac{d}{dt}(g_{\ddot{x}}\delta\dot{x}) - \left(g_{\ddot{x}} - \frac{d}{dt}g_{\ddot{x}}\right)\delta x \right] dt \quad \textcircled{2}
\end{aligned}$$

The equation ② can be multiplied and divided by $\delta v(t)$

$$\begin{aligned}
\delta^2 J &= -\frac{1}{2} \int_{t_0}^{t_f} \frac{\delta x}{\delta v} \left[\delta v \frac{d}{dt}(g_{\ddot{x}}\delta\dot{x}) - \left(g_{\ddot{x}} - \frac{d}{dt}g_{\ddot{x}}\right)\delta x \delta v \right] dt \\
&\quad \text{where } \left(g_{\ddot{x}} - \frac{d}{dt}g_{\ddot{x}}\right)\delta x \delta v = \left[\left(g_{\ddot{x}} - \frac{d}{dt}g_{\ddot{x}}\right)\delta v\right]\delta x = \left[\frac{d}{dt}(g_{\ddot{x}}\delta v)\right]\delta x \quad \text{from eq.(1)} \\
&= -\frac{1}{2} \int_{t_0}^{t_f} \frac{\delta x}{\delta v} \left[\delta v \frac{d}{dt}(g_{\ddot{x}}\delta\dot{x}) - \frac{d}{dt}(g_{\ddot{x}}\delta v)\delta x \right] dt \\
&\quad \text{where } \frac{d}{dt}(g_{\ddot{x}}\delta v\delta x - g_{\ddot{x}}\delta\dot{x}\delta v) = \frac{d}{dt}(g_{\ddot{x}}\delta v)\delta x + (g_{\ddot{x}}\delta v)\frac{d}{dt}\delta x - \frac{d}{dt}(g_{\ddot{x}}\delta\dot{x})\delta v - (g_{\ddot{x}}\delta\dot{x})\frac{d}{dt}\delta v \\
&\quad \quad \quad = \frac{d}{dt}(g_{\ddot{x}}\delta v)\delta x - \frac{d}{dt}(g_{\ddot{x}}\delta\dot{x})\delta v \\
&= \frac{1}{2} \int_{t_0}^{t_f} \frac{\delta x}{\delta v} \frac{d}{dt} [g_{\ddot{x}}(\delta v\delta x - \delta\dot{x}\delta v)] dt \\
&= \frac{1}{2} \frac{\delta x}{\delta v} g_{\ddot{x}}(\delta v\delta x - \delta\dot{x}\delta v) \Big|_{t_0}^{t_f} - \frac{1}{2} \int_{t_0}^{t_f} g_{\ddot{x}}(\delta v\delta x - \delta\dot{x}\delta v) \frac{d}{dt} \left(\frac{\delta x}{\delta v} \right) dt \\
&\quad \text{where } (\delta v\delta x - \delta\dot{x}\delta v) \frac{d}{dt} \left(\frac{\delta x}{\delta v} \right) = (\delta v\delta x - \delta\dot{x}\delta v) \frac{\delta\dot{x}\delta v - \delta x\delta\dot{v}}{(\delta v)^2} = - \left(\frac{\delta x\delta\dot{v} - \delta\dot{x}\delta v}{\delta v} \right)^2 = - \left(\delta\dot{x} - \frac{\delta x\delta\dot{v}}{\delta v} \right)^2 \\
&= \frac{1}{2} \int_{t_0}^{t_f} g_{\ddot{x}} \left(\delta\dot{x} - \frac{\delta x\delta\dot{v}}{\delta v} \right)^2 dt
\end{aligned}$$

For a necessary condition for a minimum (maximum), we need $\delta^2 J \geq 0$ ($\delta^2 J \leq 0$) for admissible $\delta x(t) \neq 0$ and $\delta x(t) \neq \delta v(t)$.

The condition for this is

$$\begin{array}{ll} \therefore g_{\ddot{x}\ddot{x}} \geq 0 & \text{for minimum} \\ g_{\ddot{x}\ddot{x}} \leq 0 & \text{for maximum} \end{array} ; \text{Legendre condition}$$

There might be finite number of times when $\left[\delta \dot{x} - \frac{\delta x \delta \dot{v}}{\delta v} \right]$ might not be finite.

Legendre condition (necessary condition) is $g_{\ddot{x}\ddot{x}} \geq 0$ (or $g_{\ddot{x}\ddot{x}} \leq 0$) along every point on the reference curve (extremal).

Legendre derived it for the scalar case; it was later extended to the vector case by Clebsch and is now known as Legendre - Clebsch conditions;

Necessary conditions:

If the extremal $\bar{x}^*(t)$ satisfies the Euler - Lagrange equation and if $g(x, \dot{x}, t)$ is such that

$$g_{\ddot{x}\ddot{x}} \geq 0 \quad (g_{\ddot{x}\ddot{x}} \leq 0) \quad ; \text{Legendre - Clebsch condition}$$

then it minimizes (maximizes) the performance index.

Note: Necessary Conditions and Sufficient Conditions

(ref.: “Optimal Control Theory for Applications”, David G. Hull, Springer, 2003, pp. 22-27)

A and B are mathematical statements. Then if the symbol “ $\rightarrow$ ” reads “implied”, and the symbol “ $\leftarrow$ ” reads “is implied by”, the following is accepted terminology:

$A \rightarrow B$ means A is sufficient for B.

$A \leftarrow B$ means A is necessary for B.

$A \leftrightarrow B$ means A is necessary and sufficient for B.

Example: Function of one variable.

It is desired to find the point x at which the function

$$I = \phi(x) \quad (i)$$

takes on an interior minimal value.

By definition, a relative minimal point is a point x which satisfies the condition

$$\Delta I = \phi(x + dx) - \phi(x) > 0 \quad (ii)$$

By expanding Eq. (ii) in a Taylor series, the following relation is obtained

$$\Delta I = dJ + \frac{1}{2} d^2 J + \dots > 0 \quad (iii)$$

$$\text{where } dJ = \phi_x dx \quad (iv)$$

$$d^2 J = \phi_{xx} dx^2 \quad (v)$$

It is well known that $\phi_x(x)$ must vanish at a minimal point as

$$\phi_x(x) = 0 \quad (vi)$$

Since $\phi_x(x) = 0 \leftarrow x$ is a minimal point,

it is a necessary condition for a minimum. The points that satisfy the equation $\phi_x(x) = 0$ are called optimal points, and they can be minima, maxima, or saddle points which are neither minima nor maxima (i.e., $\phi(x) = x^3$).

Assume that x is a solution of the optimality condition (vi). Since the first differential vanishes, the value of ΔI can be approximated by the second differential, that is

$$\Delta I \approx \frac{1}{2} d^2 J = \frac{1}{2} \phi_{xx}(x) dx^2 \quad (\text{vii})$$

From this relation, it is apparent that if

$$\phi_{xx}(x) > 0 \quad (\text{viii})$$

ΔI is positive for all $dx \neq 0$ and that x is a minimal point. Hence Eq. (viii) is a sufficient condition for a minimum because

$$\phi_{xx}(x) > 0 \rightarrow x \text{ is a minimal point.}$$

However, it is possible for $\phi_{xx}(x)$ to vanish (i.e., $\phi(x) = x^4$) and a minimal point still exist. Hence the condition

$$\phi_{xx}(x) \geq 0 \quad (\text{ix})$$

is a necessary condition for a minimum because

$$\phi_{xx}(x) \geq 0 \leftarrow x \text{ is a minimum point.}$$